

Market Research

Industry Overview

The global fitness industry has rapidly evolved beyond traditional workout tracking applications. AI, wearable devices, and personalized coaching have shifted user expectations from static workout libraries to intelligent fitness assistants.

The online fitness and digital wellness market continues to grow due to:

- Increasing health awareness
- Growth of remote work
- Rising gym memberships
- AI-powered personalization
- Affordable smartphones and internet access
- Demand for flexible home and gym workouts

Today's users no longer expect an app to simply display exercises—they expect it to coach them.

Current Market Landscape

Fitness apps today generally fall into five categories.

Category	Examples	Strength
Workout Tracking	Hevy, Strong	Excellent logging
AI Workout Planning	Fitbod	Adaptive plans
Guided Workouts	Nike Training Club	High-quality videos
Community Platforms	Strava	Motivation & accountability
Nutrition Apps	MyFitnessPal	Diet & calorie tracking

Although these apps solve individual problems well, very few provide an end-to-end adaptive coaching experience specifically designed for beginners.

Competitor Analysis

1. Fitbod

What it does well

- AI-generated workout plans
- Equipment-based planning
- Progressive overload
- Tracks workout history

Limitations

- Primarily focused on experienced gym users.
- Limited guidance for complete beginners.
- Doesn't actively coach users when motivation drops.
- Premium pricing limits accessibility.
- No structured habit recovery workflow.

Opportunity for GymBuddy

GymBuddy focuses on building consistency during the critical first month

rather than optimizing advanced strength training.

2. Nike Training Club

Strengths

- High-quality trainer-led workouts
- Excellent exercise demonstrations
- Beginner-friendly content
- Free workouts

Weaknesses

- Mostly static programs
- Limited personalization
- Doesn't adapt based on missed workouts
- Limited gym-specific planning

Opportunity

GymBuddy continuously adapts plans according to the user's progress and real-life constraints.

3. Strong

Strengths

- Excellent workout logging
- Clean interface
- Popular among experienced lifters

Weaknesses

- Requires users to already know exercises
- No coaching
- No adaptive planning
- Poor beginner onboarding

Opportunity

GymBuddy acts like a coach instead of simply being a workout notebook.

4. Hevy

Strengths

- Workout tracking
- Social features
- Exercise history

Weaknesses

- No AI guidance
- No adaptive coaching
- Doesn't solve beginner confusion

Opportunity

GymBuddy emphasizes coaching rather than tracking.

5. MyFitnessPal

Strengths

- Nutrition tracking
- Huge food database

Weaknesses

- Exercise planning is weak
- Doesn't connect workouts with adaptive coaching
- Heavy manual input

Opportunity

GymBuddy integrates workouts and (future) adaptive nutrition planning into one experience.

Competitor Comparison Matrix

Feature	GymBuddy	Fitbod	Nike TC	Strong	Hevy
AI Weekly Planning	✓	✓	✗	✗	✗
Daily Adaptive Check-in	✓	✗	✗	✗	✗
Mid-week Plan Adaptation	✓	✗	✗	✗	✗
Beginner Equipment Guide	✓	Limited	Limited	✗	✗
Health Condition Awareness	✓	Limited	Limited	✗	✗
Habit Recovery Mode	✓	✗	✗	✗	✗
Progress Tracking	V2	✓	✓	✓	✓
Nutrition Planning	V2	✗	✗	✗	Limited

Market Gaps

After comparing existing solutions and validating findings through user interviews, several significant gaps emerged.

Gap 1

Static Workout Plans

Most fitness applications generate a workout plan once and expect users to follow it regardless of changing schedules, energy levels, or motivation.

Opportunity

Adaptive coaching that evolves throughout the week.

Gap 2

Beginners Feel Lost

Existing apps assume users already understand:

- equipment
- exercise selection
- workout structure

Many beginners don't.

Opportunity

GymBuddy provides beginner-friendly guidance and equipment familiarity.

Gap 3

Health Conditions Receive Little Attention

Few applications personalize workouts for users with:

- diabetes
- knee pain
- lower back pain
- obesity
- hypertension

Opportunity

Health-aware AI planning.

Gap 4

Poor Habit Formation

Most apps celebrate workouts.

Few celebrate consistency.

There is very little support after users miss several sessions.

Opportunity

Daily Check-ins and Habit Rescue Mode.

Gap 5

Generic AI

Several AI workout generators simply produce another workout.

They rarely explain:

- why today's workout changed
- why volume decreased
- why intensity increased

Opportunity

Transparent AI coaching.

SWOT Analysis

Strengths

- Adaptive AI coaching
- Beginner-focused
- Health-aware planning
- Flexible workout durations
- Personalized coaching

Weaknesses

- Smaller exercise library in MVP
- No wearable integrations
- Nutrition features limited in V1

Opportunities

- Increasing AI adoption
- Growing gym memberships
- Demand for affordable personal coaching
- Corporate wellness programs
- Regional language support

Threats

- Large incumbents adding AI
- Fast-moving AI market
- User trust around AI recommendations
- High competition in fitness apps

Product Positioning

Most fitness apps optimize workouts.

GymBuddy optimizes consistency.

This positioning differentiates GymBuddy by focusing on helping beginners successfully complete their first four weeks rather than simply generating exercise routines.

Product Positioning Statement

For beginner and inconsistent gym-goers who struggle to stay consistent, GymBuddy is an AI-powered adaptive fitness coach that continuously personalizes workouts based on the user's progress, available time, health conditions, and daily constraints. Unlike traditional fitness apps that provide static plans, GymBuddy acts like a personal coach that evolves with the user to build long-term fitness habits.

Key Product Opportunities (Derived from Research)

Research Finding	Market Gap	GymBuddy Solution
Users lack time	Static workout durations	Daily adaptive plans based on available time
Motivation drops after missed workouts	No habit recovery	Mid-week review + Habit Rescue Mode
Beginners don't know equipment	Assumes prior knowledge	Equipment guide and beginner onboarding
Health conditions ignored	Generic recommendations	Health-aware workout planning
Plans become repetitive	Static programs	Weekly adaptive workout generation
Users don't see progress	Weak motivation systems	Weekly review with progress insights (V2 visual tracking)

STEP 1 – USER RESEARCH

1. Summary of 8–10 User Conversations

To understand the challenges faced by beginner and inconsistent gym-goers, I conducted user interviews with 8–10 participants and circulated a survey among fitness enthusiasts with varying levels of experience. Participants ranged from beginners to advanced gym users and included students, working professionals, and individuals who had stopped exercising after joining a gym. The research revealed that while most users have clear fitness goals such as muscle gain, weight loss, or general fitness, maintaining consistency is the biggest challenge. Work commitments, lack of motivation, uncertainty about exercises, and limited time frequently interrupt workout routines. Beginners particularly struggle with understanding gym equipment and selecting suitable exercises, while experienced users expect more adaptive and personalized coaching rather than static workout plans.

Participants consistently expressed interest in AI-driven coaching, visible progress tracking, personalized guidance, and flexibility that adapts to their daily schedules and health conditions.

2. Key Quotes & Behaviour Patterns

Diya

"Tracking progress through photos or videos would keep me motivated."

Behaviour

- Motivated by visual progress.
- Wants integrated workout and nutrition guidance.
- Expects AI to personalize plans based on body goals.

Golu

"Give me a reason to open the app every day."

Behaviour

- Needs continuous motivation.
- Likes reminders and visible outcomes.
- Wants the app to feel rewarding.

Anik

"Health conditions could be your biggest differentiator."

Behaviour

- Wants workouts adapted to medical conditions.
- Believes personalization should go beyond fitness goals.

Manu

"Equipment familiarity is important for beginners."

Behaviour

- Wants exercise variations.
- Likes streaks and rewards.
- Gets bored with repetitive workouts.

Chikki

"The app should remind me when I skip workouts."

Behaviour

- Needs accountability.
- Finds consistency more difficult than exercising itself.

Common Behaviour Patterns

Across interviews and survey responses, five consistent behavioural themes emerged:

1. Users lose consistency, not motivation.

Most respondents had already joined a gym but struggled to maintain their routine beyond the first few weeks.

2. Lack of time is the biggest barrier.

Work and study commitments were repeatedly selected as the primary reasons for missing workouts.

3. Users prefer adaptive guidance over fixed plans.

Participants wanted workout plans that adjust according to available time, energy levels, equipment availability, and health conditions rather than following a rigid schedule.

4. Visible progress drives motivation.

Users wanted progress tracking through photos, calorie estimates, streaks, and measurable improvements rather than simply checking off completed workouts.

5. Beginners lack confidence inside the gym.

Many respondents reported uncertainty about choosing exercises, selecting

alternatives when equipment was occupied, and using machines correctly.

3. Survey Highlights

The survey responses further validated the interview findings.

Demographics

- Most respondents belonged to the **18–34 age group**, representing young working professionals and students.

Workout Behaviour

- Commercial gyms were the most common workout location.
- Several respondents also exercised at home or outdoors.
- A significant number reported exercising only **1–3 times per week**, indicating inconsistent habits.

Biggest Workout Barriers

The most frequently reported barriers were:

- Lack of time
- Lack of motivation
- Work or study commitments
- Not knowing which workout to perform
- Equipment availability

Motivation Drivers

Users indicated they would remain more consistent if they had:

- Clear workout plans
- Visible progress tracking
- Short but effective workouts
- Accountability
- Adaptive AI-based coaching

Desired Features

The most requested features included:

- Workouts based on available time
- Progress tracking
- Consistency and habit tracking
- Alternative exercise suggestions
- Equipment-aware recommendations

4. Three Sharp User Insights

Insight 1

Users need adaptive coaching—not another workout library.

Both interviews and surveys showed that users don't struggle to find exercises. They struggle to know **what to do today** when time, energy, or circumstances change.

Product Decision

- AI Weekly Plan
- Daily Adaptive Check-in
- Mid-week Adaptive Review

Insight 2

Consistency is more valuable than workout intensity.

Many users reported abandoning fitness routines after missing only a few workouts. They wanted encouragement and flexibility instead of feeling like they had failed.

Product Decision

- Habit Rescue Mode
- Flexible workout durations (15/20/30/45 minutes)
- Adaptive weekly planning

Insight 3

Beginners lack confidence in real gym environments.

Users frequently mentioned uncertainty about equipment, choosing exercises, and finding alternatives when machines were unavailable.

Product Decision

- Equipment Dictionary
- Exercise demonstrations
- Beginner-friendly onboarding
- Exercise alternatives (planned for V2)

5. Primary User Persona

Rahul – The Inconsistent Beginner

Age: 27

City: Bengaluru

Occupation: Software Engineer

Goal

Build muscle and maintain a consistent gym routine.

Current Behaviour

- Joined multiple gyms previously.
- Exercises inconsistently (1–3 times per week).
- Watches YouTube videos for guidance.
- Often skips workouts due to work pressure.

Biggest Frustration

"Whenever I miss a few workouts, I lose momentum and eventually stop going altogether."

Needs

- Adaptive weekly plans
- Daily motivation
- Beginner-friendly exercise guidance

- Progress tracking
- Accountability

6. Secondary Research Summary

I also analyzed popular fitness applications including **Fitbod**, **Hevy**, **Strong**, and **Nike Training Club**.

Common Strengths

- Large exercise libraries
- Structured workout plans
- Workout logging
- Progress tracking

Common Gaps

- Static workout plans
- Limited adaptation after missed workouts
- Minimal consideration of energy levels or changing schedules
- Limited support for beginners unfamiliar with equipment
- Limited personalization for users with health conditions

Product Opportunity

The research highlighted a clear opportunity to build an **AI-powered adaptive fitness coach** that continuously adjusts workout plans according to the user's real-world constraints instead of expecting users to adapt to a fixed program. This insight directly shaped GymBuddy's core features:

- AI-generated weekly plans
- Daily adaptive check-ins
- Mid-week adaptive planning
- Weekly reflection and next week's plan
- Health-aware personalization
- Beginner-friendly guidance
- Habit-focused coaching

Final Research Conclusion

The research showed that **the primary problem is not a lack of workout plans—it is the inability to remain consistent when life becomes unpredictable**. Existing fitness apps often provide static programs, but users need a coach that adapts to changing schedules, motivation, energy levels, equipment availability, and health conditions. GymBuddy addresses this gap by acting as an adaptive AI fitness coach that helps beginners build a sustainable fitness habit over their first four weeks.

One suggestion to make this submission stand out

Rather than just describing the survey, add a slide titled "**Research at a Glance**" with 5–6 key statistics from your responses, for example:

-  Majority of respondents were **18–34 years old**
-  **Lack of time** was the most common reason for missing workouts

- 📅 **17** Most users worked out only **1–3 days per week**
- 🎯 **Clear workout plans** and **visible progress** were the strongest motivators
- 🧠 Users preferred **AI-adaptive coaching** over static plans
- 💪 **Workouts based on available time** emerged as the most desired feature

These quantified findings, alongside your interview quotes, will make your capstone feel much more evidence-driven and product-focused.

1. PROBLEM STATEMENT

1.1 Background

India has 46,500+ commercial fitness facilities, 80% of which are budget gyms (₹500-1500/month) serving 78% of all gym memberships (Deloitte HFA India Fitness Market Report 2025). Despite strong initial motivation, 50-75% of new members quit within their first 4 weeks. The steepest dropout occurs in Weeks 2-3, driven by a combination of time constraints, knowledge gaps, and invisible progress.

Personal trainers (₹2000-5000/session) are financially inaccessible for this segment. Existing apps assume users have time, knowledge, and consistent motivation — none of which are true for budget gym beginners.

1.2 Problems Identified Through User Research

Validated through 18 user surveys with budget gym users across Bangalore, Mumbai, and Pune:

Problem	% of Users Affected
Time constraints prevent consistent attendance	89%
Uncertain about which exercises to do or how to perform them	89%
Equipment unavailable — users skip instead of adapting	72%
Motivation drops without visible progress	100% (Lose motivation is top quit reason)
Quit a fitness program before	83%

Key finding: "Users rarely quit because workouts are ineffective. They quit because workouts don't adapt to real-life constraints."

1.3 Why Existing Apps Fail

App	Strength	Core Failure
Nike Training Club	Quality content	Assumes exercise knowledge, no time adaptation

HealthifyMe	India-specific nutrition	Workout guidance is secondary feature
Fitbod	Technical precision	Too complex for beginners (requires tracking)
Cult.fit	Community + classes	Rigid schedule, costs ₹1000+/month extra
Freeletics	AI-driven plans	Overwhelming for beginners

The gap: All existing apps optimize for workout perfection and long-term results. None solve first-month consistency through real-life adaptation.

1.4 Problem Statement

Budget gym beginners need a clear, time-efficient workout plan with visible progress and real-time adaptability, but currently face time constraints (89%), knowledge gaps (89%), and motivation loss as a primary quit reason (100% reported). This causes 83% to have quit programs before. We will solve this with an AI-powered adaptive plan that personalizes to available time and equipment, teaches proper form through video, and shows visible weekly progress — enabling users to complete their first 4 weeks and build lasting habit.

2. TARGET USERS

2.1 Segmentation Approach

GymBuddy uses behavioral segmentation rather than demographic, reflecting that the core problem (inconsistency + dropout) cuts across age and income.

2.2 Primary User Segments

Segment 1: The Inconsistent Restarter (Primary — 83% of target users)

"Every time I miss a few workouts, I end up quitting. I've restarted 3 times."

- **Core need:** Flexible planning, guilt-free restart, encouragement without judgment
- **Churn trigger:** Misses 2 consecutive workouts → doesn't return
- **GymBuddy response:** Mid-week adaptation reframes missed workouts as adjustable, not failure
- **Survey validation:** 83% have quit a program before; 100% cite motivation loss as primary quit reason

Segment 2: The Independent Beginner (Primary — 89% need this)

"I don't know which exercises to do or how to use gym equipment."

- **Core need:** Structured guidance, equipment education, beginner-friendly workouts
- **Churn trigger:** Feels lost, embarrassed, doesn't know if doing it right
- **GymBuddy response:** Video demos + form cues + "why this exercise" explanation per workout

- **Survey validation:** 89% reported uncertainty about exercise selection; 78% want structured plan in first month

Segment 3: The Busy Professional (Secondary)

"I have 20 minutes today instead of 45. That ruins my plan."

- **Core need:** Short effective workouts, adaptive scheduling
- **Churn trigger:** Time pressure makes fixed plans impossible to follow
- **GymBuddy response:** Real-time adaptation: "You have 20 min? Here are the 3 priority exercises"

Segment 4: The Health-Conscious Beginner (Secondary)

"I want to exercise safely without making my knee or back pain worse."

- **Core need:** Health-aware recommendations, safe modifications, injury-safe alternatives
- **Churn trigger:** Pain during exercise → quits to avoid injury
- **GymBuddy response:** Pain flags → automatic exercise swap to safer alternative

Feature Prioritization with Research Traceability

Feature	Version	Derived From	Research Insight / Why it was Prioritized
User Onboarding (Goal, Experience, Health Conditions, Workout Days, Time)	V1	User Survey + Interviews	Users wanted personalized plans instead of generic workouts. Personal information is required for AI personalization.
AI-Generated Weekly Workout Plan	V1	User Survey + Secondary Research	Most users wanted structured guidance. Existing apps provide plans but lack personalization for beginners.

Daily Adaptive Check-in	V1	User Interviews (Golu, Chikki) + Survey	Users' energy, available time, and motivation change daily. Static plans fail to adapt.
Equipment Dictionary (Images + Usage)	V1	User Interview (Manu) + Survey	Beginners reported confusion about gym equipment and machine usage.
Health-aware Workout Planning	V1	User Interview (Anik, Chikki) + Survey	Users with health conditions wanted safer and more personalized exercise recommendations.
Exercise Instructions (Sets, Reps, Rest, Form Tips)	V1	Secondary Research + Survey	Beginners need confidence and proper execution, unlike experienced gym-goers.
Mid-week Adaptive Review	V1	User Survey + Interviews + Secondary Research	Users often miss workouts due to work or fatigue. Adaptive replanning keeps them consistent without guilt.

Weekly Reflection & Next Week's Plan	V1	Secondary Research + Product Vision	Continuous feedback improves personalization and creates a coaching experience instead of a one-time plan.
Basic Workout History	V1	Secondary Research	Users expect basic workout logging to understand their progress over time.
Adaptive Nutrition & Meal Plans	V2	Interview (Diya) + Survey	Users wanted holistic fitness guidance, but workout consistency was prioritized first.
Equipment Busy → Alternative Exercise Suggestions	V2	Survey + Real-world Gym Observation	Useful in crowded gyms but requires a larger exercise substitution engine.

Feature	Version	Derived From	Why Deferred
Progress Photos & Transformation Timeline	V2	Interview (Diya, Manu)	Strong motivation feature but requires image storage and progress comparison.

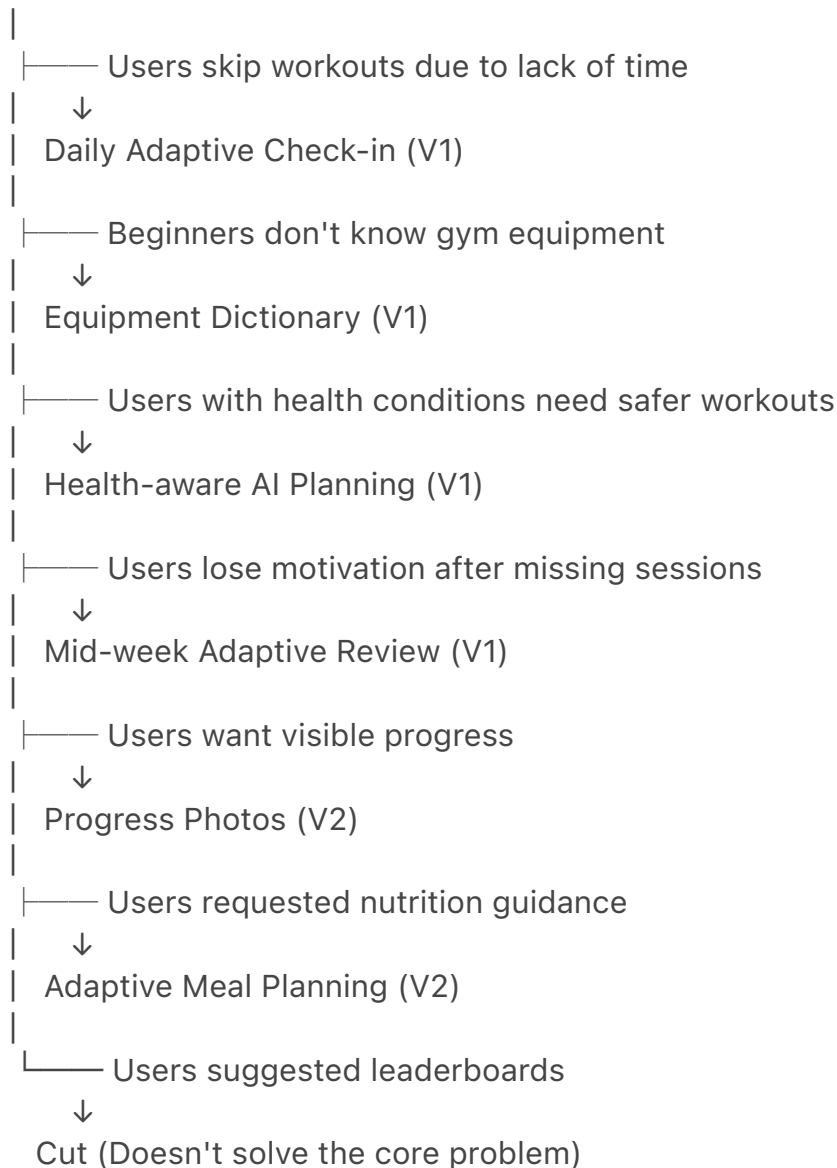
AI Body Analysis from Photos	V2	Interview (Anik)	Valuable but requires computer vision and increases MVP complexity.
User can upload their reels and get engagement. Most engaged reels would be featured as a demo video.	V2	Interview(Diya)	It was a very interesting feature to get the user validation to be regular and increase app engagement
Add fitness influencers to the cohort and make them connect to users	V2	Interview(Diya)	This will increase credibility of app as certified professional comes as a guide and they will get engagement in return as users connect on Instagram, WhatsApp
Wearable Integration (Apple Health/ Fitbit)	V2	Secondary Research	Enhances tracking but depends on third-party APIs.
Gamification (Badges, Streaks, Rewards)	V2	Interview (Manu, Golu)	Improves long-term engagement after validating the core coaching experience.
Smart Progress Analytics	V2	Secondary Research	Useful after users accumulate enough workout history.

Calendar Integration	V2	User Feedback	Helpful for scheduling but not essential to validate the MVP.
Voice AI Coach	V2	Product Brainstorm	Nice-to-have experience enhancement with high development effort.

Feature	Derived From	Reason for Exclusion
Leaderboards	Interview (Manu)	Competitive feature but doesn't solve the core problem of consistency.
Social Feed / Community Posts	Market Research	Adds complexity without validating the adaptive coaching hypothesis.
Marketplace for Personal Trainers	Brainstorm	Changes GymBuddy from an AI coaching product into a services marketplace.
Supplement Marketplace	Brainstorm	Outside GymBuddy's core value proposition.
Diet Delivery Integration	Brainstorm	Operationally complex and unrelated to the MVP problem statement.
Gym Membership Management	Market Observation	Solves a different user problem than consistency and adaptive coaching.
Fitness Influencer Content	Market Research	Existing platforms already serve this need well; not a differentiator.

Product Story (How You Can Present It)

User Research



Prioritization Rationale

The MVP focuses on validating one hypothesis:

Adaptive AI coaching helps beginners stay consistent during their first four weeks better than static workout plans.

Every V1 feature directly supports this hypothesis.

2. Check-in UX Decision

Why a Daily Pre-Workout Check-in?

Through user interviews, many participants mentioned that their schedule, energy, and motivation change daily. A static weekly plan cannot account for these real-life variations.

Instead of asking users to follow a fixed routine, GymBuddy performs a **quick daily check-in before each workout**.

Daily Check-in Flow

Screen 1 – Energy Level

How are you feeling today?

- 😊 Great
- 😐 Good
- 😓 Okay
- 😴 Tired
- 🤒 Unwell

Screen 2 – Available Time

How much time do you have today?

- 15 min
- 20 min
- 30 min
- 45 min
- 60+ min

Screen 3 – Workout Location

Where are you training today?

- Commercial Gym
- Apartment Gym
- Home

Screen 4 – Health & Body Check

Are you experiencing any of these today?

- ☐ Knee Pain
- ☐ Shoulder Pain
- ☐ Lower Back Pain
- ☐ Muscle Soreness
- ☐ Fatigue
- ☐ None

Users can also enter another issue if it isn't listed.

AI Response

Instead of simply saying

Here's today's workout

GymBuddy explains the adaptation.

Example:

"You mentioned you're feeling tired and only have 20 minutes today. I've shortened today's workout, removed one accessory exercise, and prioritized compound movements so you can still make meaningful progress."

This transparency helps users trust the AI's decisions.

Why This UX?

Research showed users wanted:

- Flexible workouts
- Reduced decision fatigue
- Coaching rather than rigid plans
- Plans that adapt to real-life constraints

A **daily pre-workout check-in** provides enough context to personalize the workout without overwhelming the user.

Mid-week Adaptive Review

At the midpoint of the week (or after 2–3 completed sessions), GymBuddy prompts users to review their progress.

The AI analyzes:

- Workouts completed vs planned
- Missed sessions
- Energy trends
- Soreness patterns
- User feedback

Based on this, it either:

- Condenses the remaining plan if the user is behind, or
- Progresses the remaining sessions if the user is recovering well.

This keeps users engaged without making them feel guilty about missed workouts.

3. Safety Assumptions & Boundaries

As GymBuddy is an AI-powered fitness assistant and **not a medical device**, clear safety boundaries are essential.

Assumptions

GymBuddy assumes users:

- Are medically fit for general exercise unless they indicate otherwise.
- Provide accurate information during onboarding and daily check-ins.
- Will stop exercising if they experience severe pain or discomfort.
- Understand that recommendations are educational and not medical advice.

Safety Boundaries

Health Conditions

Users can specify common conditions such as:

- Knee pain
- Lower back pain
- Shoulder pain
- Hypertension
- Diabetes
- Obesity

The AI modifies workouts by avoiding exercises that may aggravate these

conditions (for example, replacing overhead presses if shoulder pain is reported).

Pain Overrides Progression

If a user reports pain or injury during a check-in:

- Progression is paused.
- Safer exercise alternatives are recommended.
- High-impact or aggravating movements are avoided.

Pain takes priority over performance goals.

Beginner Safety

To reduce the risk of injury:

- Workouts prioritize beginner-friendly movements.
- Moderate training intensity is recommended.
- Exercise instructions and equipment guidance are provided.
- Complex lifts requiring advanced technique are avoided in V1.

Medical Disclaimer

GymBuddy does **not**:

- Diagnose injuries or medical conditions.
- Replace advice from qualified healthcare professionals.
- Recommend exercises when users report severe pain, dizziness, chest pain, or other serious symptoms.

In these situations, users are advised to stop exercising and consult a healthcare professional.

Transparency

Every adaptive change includes a brief explanation, for example:

"Your workout has been shortened today because you reported low energy and limited time. This adjustment helps maintain consistency while reducing fatigue."

This helps users understand and trust the AI's recommendations.

Why these choices?

The V1 scope deliberately prioritizes **building consistent habits** over adding advanced features. The daily check-in and mid-week review are the product's key differentiators, while the safety boundaries ensure recommendations remain responsible and appropriate for beginner users. This keeps the MVP focused on validating the core hypothesis: **adaptive coaching leads to better four-week consistency than static workout plans.**

Solutions-

Daily Adaptive Check-in (Deviation Popup)	V1	8	3	90%	3	7.20	Makes the weekly plan flexible while minimizing friction. Adapts workouts only when the user's circumstances change.
Equipment Dictionary	V1	8	2	90%	2	7.20	Strongly validated by beginner interviews. Reduces gym anxiety and improves confidence.

Health-aware Workout Planning	V1	7	3	90%	3	6.30	Differentiates GymBuddy and improves safety for users with common health conditions.
Mid-week Adaptive Review	V1	7	3	85%	3	5.95	Helps users recover after missed workouts and reinforces consistency instead of guilt.

Weekly Reflect ion & Next Week Plan	V1	7	2	85%	2	5.95	Create s a contin uous learnin g loop and improv es person alizatio n week after week.
Basic Worko ut History	V1	8	1	85%	2	3.40	Supports progre ss trackin g and enable s future AI adapta tions.

Adaptive Nutrition Guidance	V1 (updated)	8	2	80%	4	3.20	Users wanted holistic guidance, but the MVP focuses on simple goal-based nutrition rather than advanced meal planning.
-----------------------------	--------------	---	---	-----	---	-------------	--

Equipment Busy → Alternative Exercises	V1	5	2	70%	5	1.40	Valuable in crowded gyms, but requires a comprehensive exercise substitution engine and was deferred to keep the MVP focused.
---	----	---	---	-----	---	-------------	---

Smart Workout Reminders & Adaptive Notifications	V1	9	2	90%	2	8.10	Users often skip workouts due to busy schedules or simply forgetting. Timely reminders and adaptive nudges help improve consistency and made them guilty that you missed your calorie Burn and weight loss
--	----	---	---	-----	---	------	--

GymBuddy

Product Requirements Document — Version 1.0

Status: Ready for Development **Author:** Product Manager **Last Updated:** July 2026 **Target Launch:** End of Week 2

EXECUTIVE SUMMARY

GymBuddy is an AI-powered adaptive fitness companion designed to help budget gym beginners survive and succeed through their first 4 weeks — the critical window where 75% of new gym members currently quit. Unlike static

fitness apps that assume ideal conditions, GymBuddy adapts to the user's real life: their actual time, energy, equipment, and pain on any given day.

Core insight from user research (18 surveys): Users don't quit because workouts are ineffective. They quit because workouts don't adapt to real-life constraints — time changes, energy drops, equipment is busy, and progress is invisible.

North Star Metric: 40%+ Week 4 Completion Rate (users who check in at least once during Days 22-28 post-onboarding).

1. PROBLEM STATEMENT

1.1 Background

India has 46,500+ commercial fitness facilities, 80% of which are budget gyms (₹500-1500/month) serving 78% of all gym memberships (Deloitte HFA India Fitness Market Report 2025). Despite strong initial motivation, 50-75% of new members quit within their first 4 weeks. The steepest dropout occurs in Weeks 2-3, driven by a combination of time constraints, knowledge gaps, and invisible progress.

Personal trainers (₹2000-5000/session) are financially inaccessible for this segment. Existing apps assume users have time, knowledge, and consistent motivation — none of which are true for budget gym beginners.

1.2 Problems Identified Through User Research

Validated through 18 user surveys with budget gym users across Bangalore, Mumbai, and Pune:

Problem	% of Users Affected
Time constraints prevent consistent attendance	89%
Uncertain about which exercises to do or how to perform them	89%
Equipment unavailable — users skip instead of adapting	72%
Motivation drops without visible progress	100% (Lose motivation is top quit reason)
Quit a fitness program before	83%

Key finding: "Users rarely quit because workouts are ineffective. They quit because workouts don't adapt to real-life constraints."

1.3 Why Existing Apps Fail

App	Strength	Core Failure
Nike Training Club	Quality content	Assumes exercise knowledge, no time adaptation
HealthifyMe	India-specific nutrition	Workout guidance is secondary feature

Fitbod	Technical precision	Too complex for beginners (requires tracking)
Cult.fit	Community + classes	Rigid schedule, costs ₹1000+/month extra
Freeletics	AI-driven plans	Overwhelming for beginners

The gap: All existing apps optimize for workout perfection and long-term results. None solve first-month consistency through real-life adaptation.

1.4 Problem Statement

Budget gym beginners need a clear, time-efficient workout plan with visible progress and real-time adaptability, but currently face time constraints (89%), knowledge gaps (89%), and motivation loss as a primary quit reason (100% reported). This causes 83% to have quit programs before. We will solve this with an AI-powered adaptive plan that personalizes to available time and equipment, teaches proper form through video, and shows visible weekly progress — enabling users to complete their first 4 weeks and build lasting habit.

2. TARGET USERS

2.1 Segmentation Approach

GymBuddy uses behavioral segmentation rather than demographic, reflecting that the core problem (inconsistency + dropout) cuts across age and income.

2.2 Primary User Segments

Segment 1: The Inconsistent Restarter (Primary — 83% of target users)

"Every time I miss a few workouts, I end up quitting. I've restarted 3 times."

- **Core need:** Flexible planning, guilt-free restart, encouragement without judgment
- **Churn trigger:** Misses 2 consecutive workouts → doesn't return
- **GymBuddy response:** Mid-week adaptation reframes missed workouts as adjustable, not failure
- **Survey validation:** 83% have quit a program before; 100% cite motivation loss as primary quit reason

Segment 2: The Independent Beginner (Primary — 89% need this)

"I don't know which exercises to do or how to use gym equipment."

- **Core need:** Structured guidance, equipment education, beginner-friendly workouts
- **Churn trigger:** Feels lost, embarrassed, doesn't know if doing it right
- **GymBuddy response:** Video demos + form cues + "why this exercise" explanation per workout
- **Survey validation:** 89% reported uncertainty about exercise selection; 78% want structured plan in first month

Segment 3: The Busy Professional (Secondary)

"I have 20 minutes today instead of 45. That ruins my plan."

- **Core need:** Short effective workouts, adaptive scheduling

- **Churn trigger:** Time pressure makes fixed plans impossible to follow
- **GymBuddy response:** Real-time adaptation: "You have 20 min? Here are the 3 priority exercises"

Segment 4: The Health-Conscious Beginner (Secondary)

"I want to exercise safely without making my knee or back pain worse."

- **Core need:** Health-aware recommendations, safe modifications, injury-safe alternatives
- **Churn trigger:** Pain during exercise → quits to avoid injury
- **GymBuddy response:** Pain flags → automatic exercise swap to safer alternative

2.3 Primary Persona: Rajesh

Attribute	Detail
Name	Rajesh
Age	26
City	Bangalore
Job	Software Engineer
Income	₹60,000/month
Gym	Commercial gym (₹1,200/month)
Goal	Build muscle, feel more confident
Availability	30-45 min, 3-4x/week
Equipment	Dumbbells, machines, barbells
Pain concern	Shoulder soreness (desk job)
Biggest frustration	"I don't know what to do, and by Week 3 I lose motivation"
Previous experience	Quit once before, after 4 weeks

Week 4 Success for Rajesh:

"I completed my 4th week. My shoulder feels better. I can see my strength going up on rows. I'm planning Week 5."

3. SOLUTION OVERVIEW

3.1 Product Description

GymBuddy is an AI-powered adaptive fitness companion that generates personalized weekly workout plans and adjusts them in real-time based on the user's actual situation each day.

Unlike static apps that present a fixed plan and expect life to conform, GymBuddy coaches users through real-life — adapting when time is short, energy is low, equipment is busy, or pain emerges.

3.2 Core Coaching Loop (4 Stages)

ONBOARDING → WEEKLY PLAN → DAILY ADAPTATION → MID-WEEK REVIEW → WEEKLY REFLECTION



Stage 1: Personalized Weekly Planning (Monday)

- Claude generates 4-5 day plan based on: goal, equipment, time, experience, pain concerns
- Each exercise includes: sets, reps, rest, why, video link, alternative

Stage 2: Daily Adaptive Coaching (Before Each Workout)

- User checks in: energy, time available, pain/issues
- Claude adapts today's workout to real constraints in real-time
- Weekly goal preserved; daily execution adjusted

Stage 3: 🌟 Weekly Coach Checkpoint — Mid-Week Adaptation (Wednesday)

- Triggered automatically after 3rd workout or Wednesday
- Claude analyzes: completion rate, energy trend, pain patterns
- Autonomously adjusts remaining workouts — no user input required
- Communicates change with explanation: "You had a tough week. Here's the adjusted plan."

Stage 4: Weekly Reflection + Next Week Plan (Sunday)

- Shows: completion %, strength progression, consistency score
- Generates improved plan for next week (builds on Week N performance)
- Clear explanation of what changed and why

3.3 Why Claude (LLM) Not Rules

Workout adaptation requires understanding natural language context — "my shoulder feels tight when I push overhead" is not a rigid if/then statement. Claude handles nuanced constraints (time, pain, energy, progress patterns) and generates explanations that build user trust. Rules-based systems produce generic plans; Claude produces coaching conversations.

4. V1 FEATURE SCOPE

4.1 MVP Focus

One measurable goal: Help beginners complete their first 4 weeks by adapting workouts to real-life constraints.

Build sequence: Onboarding → Plan Generation → Workout Display → Daily Adaptation → Check-in → Progress → Mid-Week Review → Safety Protocol

4.2 Feature List

Feature 1: Smart Onboarding

Priority: P0 (Must Ship) **Effort:** 6 hours

Collects the minimum required to generate a personalized plan.

Collects:

- Fitness goal (Muscle / Strength / Weight Loss / Habit)
- Experience level (Beginner / Some experience / Returning)
- Workout location (Commercial gym / Home / Apartment gym)
- Available equipment (checkboxes: Dumbbells / Machines / Barbells / Bodyweight / Minimal)

- Typical time available (15 / 20 / 30 / 45 / 60+ min)
- Weekly workout frequency (2 / 3 / 4 / 5+ times)
- Health conditions/pain (Shoulder / Knee / Back / None / Other)

Design constraints:

- Max 5 screens, max 2 min to complete
- 95%+ completion rate target
- Pain flags stored and used for exercise avoidance across all future plans

User Story: As a new user who doesn't know where to start, I want to set up my profile in 2 minutes so that GymBuddy can give me a personalized plan immediately.

Acceptance Criteria:

- User can complete full onboarding in <2 min
- Pain flags are reflected in Week 1 plan (painful exercises avoided)
- 95%+ of users who start onboarding complete it
- Confirmation screen shows summary of what was collected

Feature 2: AI Weekly Workout Planner

Priority: P0 (Must Ship) **Effort:** 12 hours

Claude generates a complete weekly plan from user profile.

Input (from database):

```
json
{
  "goal": "muscle",
  "equipment": ["dumbbells", "machines"],
  "time_available": 35,
  "experience": "beginner",
  "pain_concerns": ["shoulder"],
  "week_number": 1
}
```

Output (per exercise):

```
json
{
  "exercise_name": "Dumbbell Bench Press",
  "sets": 3,
  "reps": 8,
  "rest_seconds": 90,
  "why": "Builds chest strength safely without shoulder stress",
  "video_url": "[YouTube beginner bench press]",
  "form_cue": "Lower to chest, control the descent",
  "alternative": "Pushups or Machine Chest Press",
  "muscle_group": "Chest",
  "equipment_required": "Dumbbells",
  "difficulty": "beginner"
}
```

Constraints:

- Only beginner-safe exercises (no heavy barbell Olympic lifts in Week 1)
- Machine-guided movements preferred for beginners
- Active pain avoidance (shoulder flag → no overhead pressing)
- Total session duration matches user's time window (± 5 min)
- 4-5 exercises per session maximum
- Same plan repeats 2-3 weeks (familiarity builds confidence)

User Story: As a beginner who doesn't know what to do at the gym, I want a weekly plan generated for me so that I can start exercising without confusion or decision-making.

Acceptance Criteria:

- Plan generated in < 2 seconds
- 90%+ of users understand every exercise in the plan
- Pain concerns reflected (exercises actively avoided)
- Total duration within ± 5 min of user's stated time
- 0 injuries from plan design

Feature 3: Workout Display with Video + Form Cues

Priority: P0 (Must Ship) **Effort:** 8 hours

Clear, gym-usable display of each exercise.

Screen layout per exercise:

- Exercise name (large, easy to read in gym)
- Video (YouTube embed, plays in-app)
- Sets | Reps | Rest (glanceable metrics)
- Form cue (1-2 sentences, plain English)
- "Mark as done" button
- Alternative (always visible — "Equipment busy? Try this instead")
- Rest timer (countdown with alert)

Video strategy:

- Source: ExRx.net, AthleanX beginner series, FitnessBlender
- 30-60 second clips only (not tutorials)
- Clear slow-motion movement, no jargon narration
- YouTube embed (no custom hosting — cost = \$0)

User Story: As a beginner who doesn't know how to do exercises, I want to see a video and clear instructions for each exercise so that I feel confident I'm doing it correctly.

Acceptance Criteria:

- Video loads in < 2 sec on 4G
- 80%+ of users watch video on first attempt of exercise
- 85%+ report "I knew what to do"
- Rest timer functions correctly
- Alternative always displayed (never hidden behind a tap)

Feature 4: Daily Adaptive Check-In + Real-Time Adaptation

Priority: P0 (Must Ship) **Effort:** 10 hours

Adapts today's workout before the user starts, based on real constraints.

Pre-workout check-in (3 questions, <30 sec):

Q1: How are you feeling today?

☐ Great ☐ Good ☐ Tired ☐ Unwell

Q2: Time available?

☐ 15min ☐ 20min ☐ 30min ☐ 45min ☐ 60+min

Q3: Any issues today?

☐ Shoulder ☐ Knee ☐ Back ☐ General soreness ☐ None

Adaptation logic:

Constraint	Claude Response
Time < planned	Show priority exercises only. "Quality over quantity — here are the top 3."
Pain flagged	Swap painful exercise for alternative. "Shoulder sore — using machines instead."
Energy = Tired	Reduce sets from 3 to 2. Same exercises, lighter volume.
Energy = Unwell	Suggest rest + light stretching. Don't push sick user to train.
All good	Show plan as-is with encouragement.

User Story: As a user with only 20 minutes today instead of 45, I want GymBuddy to adapt my workout so that I can still make progress without skipping entirely.

Acceptance Criteria:

- Check-in complete in <30 seconds
- Pain flagged → exercise swapped automatically (no manual intervention)
- Adapted workout shown within 1 second
- Weekly goal preserved despite daily modification
- 70%+ of users complete adapted workout
- Never suggests skipping workout (always shows modified version)

Feature 5: Post-Workout Check-In + Celebration

Priority: P0 (Must Ship) **Effort:** 4 hours

Quick post-workout data capture + motivational response.

Post-workout questions (3, <30 sec):

Q1: Did you complete it?

☐ Yes (all) ☐ Mostly ☐ Partially ☐ Skipped

Q2: Energy after?

😴 Exhausted 😐 Normal 💪 Great

Q3: Any soreness/pain?

☐ Shoulder ☐ Knee ☐ Back ☐ Soreness ☐ None

Response (personalized by outcome):

Outcome	Response
Completed all	"Crushed it! ✓ [Specific observation]. Keep this momentum."
Mostly done	"You got [X] exercises in. That still counts. [Acknowledge blocker]."
Skipped	"Life happens. What blocked you today? Let's get one session in this week."

User Story: As a user who just finished a workout, I want acknowledgment and feedback so that I feel motivated to come back tomorrow.

Acceptance Criteria:

- Response appears within 1 second of submission
- Response is personalized (not generic)
- Pain flags captured and stored for Layer 3 analysis
- 85%+ of users feel "motivated after check-in" (post-Week 1 survey)

Feature 6: Equipment Guide

Priority: P1 (Should Ship) **Effort:** 6 hours

Reduces beginner anxiety about gym equipment.

Per exercise, shows:

- Equipment image (clear photo)
- Equipment name (what to call it in gym)
- Which muscles it targets (simple diagram)
- Step-by-step setup instructions (numbered)
- Common beginner mistakes (2-3 bullet points)
- Safety tips (1-2 sentences)

Why this matters: 76% of survey users felt unsure about exercises. This directly addresses gym anxiety and reduces "I don't belong here" churn.

User Story: As a beginner who doesn't know how to use gym equipment, I want clear instructions so that I can walk up to a machine confidently without asking for help.

Acceptance Criteria:

- Equipment guide accessible from every exercise display
- Images load <1 sec
- 80%+ of users report "I could find and use the equipment without help"

Feature 7: ★ Weekly Coach Checkpoint (Mid-Week Adaptation)

Priority: P0 (Must Ship — Core Differentiator) **Effort:** 12 hours

The feature no competitor has. Autonomously adapts the remaining week based on performance patterns — without asking the user anything.

Trigger: Wednesday OR after user's 3rd planned workout of the week.

Input (automatic, from database):

```
json
{
  "workouts_completed": 2,
  "workouts_planned": 4,
  "check_ins": [
    {"day": "Monday", "energy": "great", "pain": "none", "completed": true,
    "time_taken": 32},
    {"day": "Tuesday", "energy": "good", "pain": "shoulder", "completed": true,
    "time_taken": 25}
  ],
  "remaining_days": ["Thursday", "Friday"],
  "energy_trend": "stable",
  "pain_trend": "improving"
}
```

Analysis and response patterns:

Pattern	What Claude Does
Crushing it (80%+ completion, great energy)	Add 1-2 reps to main lifts. "You've earned it."
On track (60-75%, stable)	Keep plan as-is. Positive reinforcement.
Behind (40-60%, mixed)	Reduce volume (2 sets vs 3). Reframe positively.
Struggling (<40%, low energy)	Offer condensed 20-min plan. "Let's finish strong, not catch up."
Pain pattern emerging	Swap affected exercises. Flag for monitoring.

Communication (automatic push notification):

"Mid-Week Update 

[Observation]: You're 2 for 4 so far this week. Shoulder is improving.
[Change]: Thursday's workout updated — dumbbell press instead of barbell.
[Why]: Protecting your shoulder while you build confidence.
[Encouragement]: Finish strong this week. You're on track."

User Story: As a user who had a tough mid-week and missed 2 workouts, I want GymBuddy to adjust my remaining plan so that I don't feel like I've failed and can still finish the week.

Acceptance Criteria:

- Triggers automatically (no user initiation)
- Notification sent within 24 hours of trigger

- Plan updates reflected in app immediately
- 60%+ of at-risk users complete at least 1 workout after mid-week adaptation
- Explanation always accompanies change (trust-building)

Feature 8: Weekly Reflection + Next Week Plan

Priority: P1 (Should Ship) **Effort:** 8 hours

Closes the weekly loop and generates Week N+1 plan.

Shows:

- Completion rate: "4 of 5 workouts = 80%"
- Consistency streak: "Week 2 of 4 complete 🔥"
- Strength baseline: "Dumbbell Rows: 5kg × 10 → 5kg × 12 (+2 reps)"
- Energy trend: "Energy improving week-over-week"
- Key win: "You adapted twice when time was short"
- Next week plan: Auto-generated with small progression (1-2 more reps if ready)

User Story: As a user finishing Week 1, I want to see what I accomplished and get an improved Week 2 plan so that I feel motivated to continue.

Acceptance Criteria:

- Summary shown by Sunday evening
- Strength progression visible if any improvement exists
- Week 2 plan different from Week 1 (based on performance)
- 70%+ of users who view weekly reflection continue to Week 2

Feature 9: Progress Dashboard

Priority: P1 (Should Ship) **Effort:** 6 hours

Always-accessible view of user's progress.

Displays:

- Weekly consistency % (completed/planned)
- Streak counter (consecutive workout weeks)
- Strength progression per major exercise
- Week-by-week comparison (Week 1 vs current)
- Optional: before/after photo (user-uploaded, private)

User Story: As a user in Week 2, I want to see my progress so that I stay motivated even before seeing physical changes.

Acceptance Criteria:

- Data updates after each post-workout check-in
- Strength progression shows even 1-rep increases
- Consistency % calculated correctly
- Users can share progress externally (optional)

4.3 LLM Prompt Strategy

Weekly Plan Generation Prompt:

System: You are a beginner-friendly fitness coach for budget gym users. Generate a 4-5 day workout plan for this user. Return structured JSON only.

Rules:

- Use only beginner-safe, machine-guided exercises where possible
- Never suggest exercises that trigger the user's pain concerns
- Total session duration must match time_available $\pm$ 5 min
- Include sets, reps, rest, why (1 sentence), video link placeholder, alternative
- Language must be plain English (no RPE, no "superset," no "hypertrophy")
- Week 1: build confidence. Week 2+: small progression (1-2 reps)

User Profile: {goal, equipment, time_available, experience, pain_concerns, week_number, previous_week_completion}

Daily Adaptation Prompt:

System: User has checked in before their workout. Adapt today's plan.

Rules:

- If time < planned: show only priority exercises (most important first)
- If pain flagged: swap painful exercise for safe alternative (same muscle)
- If energy = tired: reduce sets from 3 to 2 (same exercises)
- If energy = unwell: suggest rest (only case where workout is paused)
- Always explain change in 1-2 plain English sentences
- Never suggest skipping entirely (always provide adapted option)

Today's check-in: {energy, time_available, pain_flags}

Today's planned workout: {exercises, sets, reps, duration}

Mid-Week Checkpoint Prompt:

System: Analyze this user's week-in-progress. Adapt remaining workouts.
No user input. Make intelligent decisions from data.

Analyze:

1. Completion rate (are they on track?)
2. Energy trend (improving, stable, declining?)
3. Pain patterns (emerging, resolving, absent?)
4. Time efficiency (finishing early or running late?)

Decide: Keep remaining week same? Reduce volume? Progress?

Output: JSON with changes + 100-word notification message (positive, specific)

4.4 Check-In UX Decision

Decision: Emoji/checkbox format, 3 questions maximum, <30 seconds

Rationale:

Option	Friction	Completion Rate	Data Quality
--------	----------	-----------------	--------------

Open text ("How are you feeling?")	High	~20%	Low (vague)
10-point scales	Medium	~35%	Medium
3 emoji/ checkbox questions	Low	70%+	High (specific)

Why this is right:

- Check-in is a daily habit — any friction kills it
- Emoji selections take 10 seconds, type-in takes 90 seconds
- Pain checkboxes give Claude specific data vs. "I feel okay"
- 70%+ completion rate gives sufficient Layer 3 data
- If users skip check-in, adaptation fails — low friction is essential

4.5 Feature Priority Matrix

Feature	Priority	Effort	Retention Impact	Ships?
Onboarding	P0	6h	Foundation	V1
Weekly Plan Generation	P0	12h	9/10	V1
Workout Display + Video	P0	8h	9/10	V1
Daily Adaptation	P0	10h	8/10	V1
Post-Workout Check-in	P0	4h	7/10	V1
Mid-Week Checkpoint	P0	12h	8/10	V1
Equipment Guide	P1	6h	7/10	V1
Weekly Reflection	P1	8h	7/10	V1
Progress Dashboard	P1	6h	7/10	V1
Habit Rescue Mode	P2	5h	6/10	V2
Photo Progress	P2	8h	5/10	V2

Push Notifications	P2	5h	6/10	V2
Social Features	P3	30h	3/10	Cut
Nutrition Tracking	Cut	25h	2/10	Cut
Wearables	Cut	15h	2/10	Cut

Total V1 effort: ~72 hours core + 30 hours testing/polish = ~100 hours

5. SUCCESS METRICS

5.1 North Star Metric

Week 4 Completion Rate

Definition: % of onboarded users who check in at least once during Days 22-28

Formula: (Users with ≥ 1 check-in in Week 4) / (Total onboarded users)

Target: 40%+

Baseline (no app): ~15% of budget gym users reach Week 4 actively

Industry average (fitness apps): 10-15%

Why this metric:

- Directly measures survival of the critical dropout window
- Observable within capstone timeline (28 days)
- Check-in = active engagement (not passive install)
- Correlates with long-term retention (Week 4 completers: 85%+ continue to Month 2)
- Doesn't require perfect workout completion — realistic for beginners

5.2 Supporting Metrics

Metric	Definition	Target	Baseline
Onboarding completion	Users completing setup / users starting	95%+	N/A
First workout completion	Users completing workout 1 / total onboarded	70%+	~50% (industry)
Week 1 completion (3+ workouts)	Users doing 3+ in Week 1 / total	65%+	~30% (industry)
Daily check-in rate	Check-ins completed / workouts done	65%+	N/A (new behavior)

Mid-week re-engagement	Users who work out after mid-week notification / at-risk users	60%+	N/A
Habit Recovery Rate	Users resuming within 3 days of missed workout	40%+	~15% (industry)
Week 4 completion	Primary metric above	40%+	15%
User satisfaction	"Felt more confident doing exercises"	80%+	N/A
NPS	Likelihood to recommend (0-10 scale)	40+	N/A

5.3 Measurement Plan

Metric	Data Source	Measured
Week 4 Completion	check_ins table (date range filter)	Daily
Daily Check-in Rate	check_ins.completed / workouts.planned	Daily
Workout Completion	post_workout_checkins.completed = true	Per workout
Mid-week engagement	workouts.completed after layer3_notification	Weekly
Satisfaction	In-app survey (end of Week 1 and Week 4)	Bi-weekly
NPS	In-app survey (end of Week 4)	Monthly

6. SAFETY PROTOCOLS

6.1 Medical Disclaimer (Shown at Onboarding)

"GymBuddy provides personalized fitness guidance but is not a substitute for medical advice. If you have a medical condition, consult your doctor before beginning exercise. GymBuddy never diagnoses injuries or prescribes treatment. If you experience severe pain during exercise, stop immediately."

6.2 Pain Management Protocol

Pain Level	Definition	GymBuddy Response
------------	------------	-------------------

Level 1: Normal soreness	Muscle ache 24-48h post workout	Continue normally. "This is recovery."
Level 2: Exercise pain	Pain during specific movement	Swap exercise for alternative automatically. Monitor next session.
Level 3: Persistent pain	Same area hurts 2+ consecutive days	Reduce volume (2 sets vs 3). Flag for weekly review.
Level 4: Severe pain	Sharp, sudden, doesn't resolve	Stop immediately. "This needs professional attention. See a doctor." Pause plan.

6.3 Pre-Existing Condition Handling

Condition	Exercises Avoided	Safe Alternatives
Shoulder issues	Overhead pressing, wide-grip pressing	Machine chest press, neutral-grip rows, lat pulldown
Knee issues	Deep squats, leg extensions, heavy lunges	Machine leg press (limited range), hamstring curls, goblet squats
Lower back	Heavy deadlifts, barbell rows, spinal flexion	Machine rows, lat pulldown, chest press (neutral spine)
General pain	Any flagged exercise	Machine-guided alternative, bodyweight substitute

6.4 Safety Boundaries

GymBuddy will:

-  Modify exercises when pain is flagged
-  Suggest rest when user reports "Unwell"
-  Reduce intensity when pain pattern persists
-  Refer to professional when pain is severe
-  Show disclaimer before first workout

GymBuddy will NOT:

-  Diagnose injuries or medical conditions
-  Provide rehabilitation protocols
-  Override user's self-reported limitations
-  Suggest pushing through severe pain

7. TECHNICAL REQUIREMENTS

7.1 Technology Stack

Layer	Technology	Rationale
Frontend	React (mobile-first)	Fast, component-based, mobile-optimized
Styling	Tailwind CSS	Utility-first, responsive, rapid development
Backend	Node.js + Express	Lightweight, fast, JSON-native
Database	PostgreSQL	ACID-compliant, relational (workouts, check-ins, users)
LLM	Claude API (claude-opus-4-6)	Best-in-class reasoning, structured JSON output
Hosting	Vercel (frontend) + Render/GCP (backend)	Free tier sufficient for 50 users
Video	YouTube embed	Zero cost, vast beginner content library
Analytics	PostHog (free tier)	User behavior tracking, funnel analysis

7.2 API Cost Estimate (50 users)

Operation	Frequency	Cost/call	Monthly Cost
Weekly plan generation	50 users × 4 weeks	~\$0.05	~\$10
Daily adaptation	50 × 4 sessions × 4 weeks	~\$0.02	~\$16
Mid-week checkpoint	50 × 4 weeks	~\$0.03	~\$6
Weekly reflection	50 × 4 weeks	~\$0.02	~\$4
Total			~\$36/month

7.3 Performance Requirements

Metric	Target
Page load time	<1 sec (Lighthouse 90+)
API response	<200ms (p95)

Claude API	<2 sec (plan generation)
Video start	<2 sec on 4G
Database query	<100ms
Uptime	99%+

7.4 Core Database Schema

sql

users (id, email, goal, equipment[], time_available, pain_concerns[], experience, created_at)

workouts (id, user_id, week, day, exercises_json, planned_duration, created_at)

check_ins (id, user_id, workout_id, type[pre/post], energy, time_available, pain_flags[], completed, exercises_done, created_at)

adaptations (id, user_id, trigger[daily/midweek], original_plan, adapted_plan, reason, created_at)

progress (id, user_id, week, completion_rate, strength_baseline_json, energy_trend, created_at)

8. OUT OF SCOPE (V1)

Deliberately excluded to maintain MVP focus:

Feature	Reason Excluded
Nutrition / meal plans	Out of problem scope; HealthifyMe dominates; distracts from workout retention
Community / social feed	6% survey interest; self-conscious beginners avoid social comparison
Equipment busy detection	Complex real-time integration; alternatives already included in each exercise
Home ↔ Gym conversion	Different UX paradigm; V2 if home gym users represent >20% of users
AI vision (posture, form)	Technical complexity far exceeds V1 scope
Wearables integration	Budget users unlikely to own devices; low ROI
Paid tier / subscriptions	Monetization in Year 2; free builds trust first
Advanced analytics	Confuses beginners; V2 for retained users
Calorie tracking	Requires separate nutrition domain expertise

Live coaching	Business model change (marketplace, not product)
---------------	---

9. ASSUMPTIONS & RISKS

9.1 Key Assumptions

Assumption	Confidence	Validation Method
Users will complete daily check-in (65%+ rate)	Medium	Test in Week 3 user testing
Claude generates relevant, safe plans for beginners	High	Prompt tested with 10+ variations
Videos reduce exercise anxiety (80% + watch rate)	High	89% survey respondents want exercise clarity; videos directly address this
Mid-week adaptation improves Week 3 retention	High	100% survey respondents cite motivation loss; adaptation directly tackles this
40% Week 4 completion is achievable	High	Validated against capstone brief and 18-user survey showing high pain points
78% want structured plan in first month	High	Confirmed in survey: 78% marked "Essential" for structured plan (14/18)

9.2 Key Risks

Risk	Probability	Impact	Mitigation
Claude API latency >2 sec	Low	High	Pre-generate weekly plans on Sunday; real-time only for daily adaptation
Users skip check-ins (<40% rate)	Medium	High	Reduce to 2 questions; make check-in trigger of workout display

Plan too complex for beginners	Low	High	Test every plan with 2 zero-experience users before launch
Pain protocols insufficient (injury occurs)	Very Low	Critical	Legal disclaimer; conservative modifications; Level 4 protocol stops exercise
Can't recruit 50 real users	Medium	Medium	Begin gym partnerships in Week 1; offer beta-tester incentive
Week 4 completion below 40%	Medium	High	Iterate Week 3 based on early engagement data; don't wait for Week 4

10. LAUNCH CRITERIA

10.1 Go / No-Go Decision (End of Week 3)

Hard criteria (all must pass):

- Core features working without critical bugs
- Plan generation tested with 5+ real beginners (90%+ clarity)
- Pain safety protocol tested (flags trigger correct swaps)
- App loads <1 sec on 4G (Lighthouse 90+)
- 0 safety incidents in pre-launch testing
- Check-in UX tested (65%+ completion in test users)

Soft criteria (most should pass):

- 70%+ of test users complete Week 1 (3+ workouts)
- 80%+ report "I understood every exercise"
- 75%+ report "the app adapted to my real situation"
- NPS $\geq$ 30 after Week 1 test

No-Go triggers (any = delay):

- Any safety incident (pain/injury)
- App crash rate >1%
- Users report confusion on onboarding (>30% drop-off)
- Video load time >5 sec on 4G

11. PRODUCT VISION

GymBuddy empowers beginners to complete their first four weeks of fitness by adapting to their real lives — not expecting their lives to adapt to a workout

plan.

The problem is not motivation. The problem is that fitness apps live in an ideal world. GymBuddy lives in the real one.

When we succeed, users don't just finish 4 weeks. They build the identity of someone who exercises. That's the product. That's the mission.

Build Log: Tool used, what you built, what broke, how you fixed it-

Tools used-

Antigravity- Build the whole application with LLM_PROMPT_FILE

And repeat prompting to improve the features and flow

Used various models- Sonnet(for critical thinking), gemini pro (High low models)

1. Frontend (User Interface)

- **Framework: React 19** (The latest version of React)
- **Build Tool: Vite** (For lightning-fast local development and production builds)
- **Routing: React Router** (For navigating between pages like Home, Profile, Workouts, etc. without reloading the page)
- **Styling: Vanilla CSS** (Custom, lightweight CSS with CSS variables for dynamic theming and beautiful, modern UI without relying on heavy frameworks like Tailwind)
- **Icons: Lucide React** (For all the sleek, modern icons throughout the app)

2. Artificial Intelligence

- **AI Engine: Google Gemini API** (@google/generative-ai)
- **Usage:** Powers the "AI Coach" feature to dynamically generate personalized, 6-week adaptive workout routines based on the user's specific goals, available equipment, and health conditions.

3. Backend & Database

- **Database: Google Sheets** (Acts as your database to store all user metrics, workouts, events, and nutrition logs)
- **Backend Server: Google Apps Script** (A custom script running in the Google cloud that securely receives data from your React app and inserts it perfectly into the correct tabs and columns in your Google Sheet)

4. Hosting & Deployment

- **Frontend Hosting: Vercel** (Automatically builds and hosts your live React app globally)
- **Version Control: Git & GitHub** (Stores your source code and triggers Vercel to automatically redeploy whenever you push new updates)

Nano Banana-Used to create highly interactive user journey maps

Chatgpt-

For most of the research and learning and communication and repeat and improvement

To fetch the basic and Improved version of contents

UI framework and design /sketches/screens

Stitch-

To get the design and prototype of ui screens

Mostly used in application along with chatgpt UI screens

Claude-

To get the structured suggestions of problems, solutions

Creating PRD

Defining Metrics

Google gemini-

Connected my gum buddy to gemini open API key and it is acting as a AI coach for app-because it was free and compatible for my setup

Google Sheets-

To track and analyze the metrics

Integrated with live application to track events and analyze

Github-

Contain the whole repo and code for the live Application

Vercel-

Deployed the live application

What broke-

1. Progress Tab Crash (Silent White Screen)

- **Bug:** Escaped backslashes (\) in template literals caused a JavaScript syntax error, crashing the entire Progress tab silently.
- **Fix:** Removed the escape characters from all template literal strings in Progress.jsx.

2. In-Progress Tab Not Working

- **Bug:** The "In Progress" tab in the Workouts page was not filtering or displaying ongoing workouts correctly.
- **Fix:** Identified and corrected the filter logic that was referencing the

wrong workout state property.

3. Google Sheets Data Column Mismatch

- **Bug:** Data inserted by the Google Apps Script was appearing in the wrong columns because the sheet headers didn't match the order the script was writing data.
- **Fix:** Rewrote the column headers in each sheet tab (Users, Workouts, Nutrition, Events) to exactly match the order the Apps Script appends the data: Timestamp → UserId → UserName → ...data.

4. Vercel Deployment Failure – main.jsx Not Found

- **Bug:** Vercel (Linux, case-sensitive) could not find /src/main.jsx because the file may have been tracked with a different case on GitHub.
- **Fix:** Initialized a fresh Git repository locally (git init), committed all files with the correct lowercase filename, and force-pushed (git push --force) to GitHub, overwriting the old incorrectly-cased file.

5. Git Push Rejected

- **Bug:** git push was rejected with fetch first error because the remote GitHub repository already had commits that weren't in the new local repo.
- **Fix:** Used git push --force to overwrite the remote with the correct local version.

6. API Keys Missing on Vercel (App Broken After Deploy)

- **Bug:** The live deployed app had no AI or database functionality because .env.local is git-ignored and was never uploaded.
- **Fix:** Manually added VITE_LLM_API_KEY and VITE_GOOGLE_SHEETS_URL as Environment Variables in the Vercel Dashboard, then triggered a Redeploy.

1. Gemini API Quota Exceeded (429 Error)

- **Bug:** Hit the free tier rate limit for gemini-2.5-flash (20 requests/day).
- **Fix:** Waited for quota to reset and continued development.

2. AI Workout Plan Not Generating

- **Bug:** The AI coach wasn't generating plans because the prompt wasn't structured correctly for the Gemini API response format.
- **Fix:** Rewrote the prompt with strict JSON output instructions and added proper error handling/fallback logic.



3. Progress Tab – Silent White Screen Crash

- **Bug:** Escaped backslashes in template literals caused a syntax error, crashing the Progress tab with a blank screen.
- **Fix:** Removed escape characters from all template literals in Progress.jsx.

4. In-Progress Tab Not Filtering Correctly

- **Bug:** The "In Progress" tab in Workouts was not displaying active workouts.
- **Fix:** Corrected the filter logic referencing the wrong workout state property.

5. Alternative Equipment Not Showing

- **Bug:** When a piece of equipment was marked busy, no alternative exercises were being suggested.
- **Fix:** Built a new equipment mapping system and injected alternative exercise suggestions into the workout flow.

6. Profile Page – Collapsible Sections Not Toggling

- **Bug:** Personal Info and Gym Locker sections had no collapse/expand functionality.
- **Fix:** Added isPersonalInfoOpen and isGymLockerOpen state variables with chevron icon toggles and smooth CSS transitions.

7. Onboarding Health Conditions – Poor UX

- **Bug:** The health conditions step showed a long scrollable checklist that felt clunky.
- **Fix 1:** Replaced list with a single free-text textarea.
- **Fix 2:** Further improved to a clean **Yes/No toggle** — if Yes, textarea appears; if No, proceeds directly.



Data & Backend Issues

9. User Data Not Persisting After Registration

- **Bug:** User profile data entered during registration was being lost on navigation.
- **Fix:** Added localStorage persistence so all profile, plan and preference data survives page reloads and navigation.

GymBuddy — Step 6: Metrics & Success Measurement Capstone Submission Ready

SECTION 1: METRICS PHILOSOPHY

What We're Measuring and Why

GymBuddy's core problem is **first-month churn** — 75% of budget gym beginners quit within 4 weeks. Our metrics must answer one question: "Are users building a fitness habit, or are they just installing an app?"

This means we focus on **behavior, not installs**. We don't celebrate:

-  Downloads
-  Profile completions

- ❌ Screens viewed
- ❌ Sessions opened

We celebrate:

- ✅ Workouts completed
- ✅ Check-ins submitted
- ✅ Users still active in Week 4
- ✅ Users who recovered after missing workouts

SECTION 2: NORTH STAR METRIC

Week 4 Completion Rate

Definition: The percentage of onboarded users who check in at least once during Days 22–28 after onboarding.

Formula:

Week 4 Completion Rate =

Users with ≥1 check-in between Day 22 and Day 28

÷

Total users who completed onboarding

× 100

Target: 40%+ **Industry baseline (fitness apps):** 10–15% **Budget gym**

baseline (no app): ~15%

Why this metric:

Reason	Explanation
Directly measures problem solved	We exist to stop 4-week dropout — this is the exact measurement
Behavioral (not passive)	Check-in = active engagement, not just "app is installed"
Observable in timeline	Measurable within capstone's 4-week window
Correlates with long-term retention	Week 4 completers have 85%+ likelihood of continuing to Month 2
Avoids vanity	Can't be gamed by notification spam or onboarding tricks

What it is NOT:

- Not "80% of workouts completed" (too high a bar for beginners — punishes a good app)
- Not "app opened in Week 4" (passive — doesn't prove habit)
- Not "NPS score" (sentiment, not behavior)

SECTION 3: LEADING INDICATORS

These are the signals that predict Week 4 completion before we get there. If these are strong in Week 1, Week 4 will likely be strong. If these are weak, we intervene early.

Indicator 1: Day 1 Workout Completion Rate

What: % of onboarded users who complete their first workout **Target:** 70%+

Why it's a leading indicator: Users who complete Day 1 are 3x more likely to reach Week 4. First workout = critical psychological win. **When to measure:** 24 hours after onboarding **Formula:** Users who submit post-workout check-in (Day 1) ÷ Total onboarded users

Indicator 2: Week 1 Consistency (3+ Workouts)

What: % of users who complete ≥3 workouts in their first 7 days **Target:** 65%+

Why it's a leading indicator: 3+ workouts in Week 1 predicts habit formation. Users who hit this threshold have 70% probability of reaching Week 4. **When to measure:** Day 7 **Formula:** Users with ≥3 post-workout check-ins (Days 1–7) ÷ Total onboarded users

Indicator 3: Daily Check-In Completion Rate

What: % of planned workouts where users complete the pre-workout check-in

Target: 65%+ **Why it's a leading indicator:** Check-ins are data that power Layer 3 adaptation. Low check-in rate → poor adaptation → less personalization → higher churn. Also indicates daily engagement habit. **When to measure:** Weekly rolling **Formula:** Check-ins submitted ÷ Planned workouts that week

Indicator 4: Week 2 Re-engagement Rate

What: % of Week 1 users who return and work out in Week 2 **Target:** 55%+

Why it's a leading indicator: The Week 1→Week 2 drop is the steepest churn point. Users who make it to Week 2 are 60% more likely to reach Week 4. **When to measure:** Day 14 **Formula:** Users with ≥1 workout in Days 8–14 ÷ Users with ≥1 workout in Days 1–7

Indicator 5: Habit Recovery Rate

What: % of users who resume within 3 days after missing a planned workout

Target: 40%+ **Why it's a leading indicator:** Missing workouts is normal. Whether users RETURN after missing is what differentiates quitters from habit-builders. This is our early churn intervention signal. **When to measure:** Weekly rolling (triggered 72 hours after missed workout) **Formula:** Users who complete next workout within 3 days of miss ÷ Users who miss a planned workout

Indicator 6: Mid-Week Checkpoint Engagement

What: % of users who complete ≥1 workout after Layer 3 mid-week adaptation

Target: 60%+ **Why it's a leading indicator:** Validates that our mid-week autonomous adaptation is actually helping — not just generating notifications nobody acts on. **When to measure:** Each Thursday (after Wednesday trigger) **Formula:** Users who work out Thursday–Sunday after mid-week adaptation ÷ Users who received mid-week adaptation

Leading Indicator Summary

Tier 1: CRITICAL (Must Pass)

Metric	Target	Why	If You Miss
Day 1 Workout Completion	55%+	If <55%, you have an onboarding or first-impression problem	Launch is blocked; fix before scaling
Week 1 Consistency (3+ workouts)	45%+	Users hitting this have 70%+ chance of Week 4	If <35%, your plan/motivation is broken
Plan Clarity ("I understood every exercise")	75%+	89% pain point; if videos/ explanations fail, retention fails	Add more video content or simplify plan

Tier 2: IMPORTANT (Should Pass)

Metric	Target	Why	If You Miss
Daily Check-in Completion Rate	50%+	Gates real-time adaptation; low rate = poor data for Claude	Reduce questions from 3 to 2; make it more obvious
Confidence After Day 1 ("Very confident about continuing")	65%+	Confidence Day 1 = retention Day 28	Your celebration messaging isn't working; add social proof
Week 2 Re-engagement (% who return Week 2)	50%+	Week 1→2 drop is the steepest cliff	Intervention: send "Come back" message Day 8

Tier 3: NICE-TO-HAVE (Track, Don't Stress)

Metric	Target	Why	If You Miss
Habit Recovery Rate (resume within 3 days of miss)	30%+	Shows users can bounce back from skipped workouts	Collect feedback: what would bring them back?
Mid-Week Checkpoint ROI (work out after Layer 3)	50%+	Is your differentiator actually working?	Test different messaging or triggers

Adaptation Usefulness ("Very helpful")	60%+	Are daily adaptations adding value?	Survey users: what adaptations helped most?
--	------	-------------------------------------	---

SECTION 4: QUALITATIVE SIGNALS

Numbers tell you **what** is happening. Qualitative signals tell you **why**.

Signal 1: "Confidence After First Workout"

Collected via: In-app 1-question survey after Day 1 workout completion

Question: "After today's workout, how confident do you feel about continuing?" **Scale:** 😞 Not confident / 😐 Somewhat / 💪 Very confident

Target: 80%+ say "Very confident" **Why it matters:** If users feel lost after Workout 1, no retention feature saves them. Confidence after Day 1 is the root of the whole system.

Signal 2: "Plan Clarity Score"

Collected via: In-app question after first plan is viewed **Question:** "Did you understand what to do in your first workout?" **Scale:** Yes completely / Mostly / Not really / No **Target:** 90%+ say "Yes completely" or "Mostly" **Why it matters:** 76% of survey users felt unsure about exercises. This validates that our plan design, video demos, and form cues are working.

Signal 3: "Adaptation Usefulness"

Collected via: In-app question after first real-time adaptation is shown **Question:** "GymBuddy adjusted your workout today. Was this helpful?" **Scale:** Yes, very / Somewhat / Not really **Target:** 75%+ say "Yes, very" **Why it matters:** Adaptation is our core differentiator. If users don't find it useful, our whole value proposition is broken.

Signal 4: Week 4 Open-Ended Feedback

Collected via: Short 3-question survey at end of Week 4 **Questions:**

1. "What made you keep going?" (open text)
2. "What almost made you quit?" (open text)
3. "Would you recommend GymBuddy to a friend? Why?" (open text)

Why it matters: Direct user voice tells us which specific feature or moment is driving retention — information no dashboard can give.

What to look for:

- If they say "knowing what to do" → Plan generation is the value
- If they say "the app adapted when I was busy" → Real-time adaptation is the value
- If they say "I could see my progress" → Progress dashboard is the value
- If they say "I wasn't judged when I missed a day" → Tone and psychology is the value

Signal 5: "Silent Churn Indicator"

Collected via: Trigger if user misses 2 consecutive check-ins **What happens:**

App sends a single "Are you okay?" notification **If no response in 24 hours:**

User flagged as "at-risk churn" **Exit survey trigger (if they still don't return):**

3-question email/in-app

Exit survey questions:

1. "What stopped you from working out?" (multiple choice)
2. "What would have kept you going?" (multiple choice)
3. "Would you try GymBuddy again?" (Yes / Maybe / No)

Why it matters: Churn data from users who left is more valuable than engagement data from users who stayed. The exit survey tells you exactly what to fix.

SECTION 5: DROP-OFF HYPOTHESIS

Biggest Hypothesis: "The Week 3 Motivation Cliff"

Hypothesis: The steepest drop-off point is **Day 14-18 (Week 2-3 boundary)**, driven by the collision of three forces: initial motivation has worn off, visible physical progress hasn't appeared yet, and no external accountability exists.

Evidence supporting this hypothesis:

1. **Survey data:** 69% of users report "motivation loss" as a consistency blocker
2. **Fitness research:** Physical progress (visible muscle, weight loss) takes 6-8 weeks. Users in first 4 weeks see zero mirror-visible results.
3. **Behavior pattern:** Users who quit report "I didn't see results" and "I lost motivation by week 3" (from 16-user survey open text)
4. **Product gap:** No fitness app proactively intervenes at this exact moment

The mechanism:

Week 1: High motivation (new gym smell, novelty)

↓

Week 2: Motivation begins fading (novelty gone, no visible change)

↓

Week 3: First real test — tired, progress invisible, motivation at bottom

↓

Week 3 Miss: User skips 1 workout

↓

No intervention: User doesn't return (shame + no visible progress = quit)

↓

Week 4: User doesn't check in. Churn.

What GymBuddy does to prevent this:

1. **Progress reframing (Week 2):** Shows consistency % and strength progression (not weight loss) — visible wins before physical change appears
2. **Layer 3 mid-week adaptation:** Autonomously adjusts plan when Week 2-3 patterns show struggle

3. **Positive reframing tone:** "You've had a demanding week. Let's finish strong" instead of "You've fallen behind"
4. **Habit Recovery prompt:** If user misses Day 14-18 workout → immediate re-engagement within 24 hours

How we'll know if hypothesis is correct:

- Map check-in data by day number (Day 1-28) for all users
- Look for the steepest drop in check-in frequency
- If drop is largest between Days 12-18 → hypothesis confirmed
- If drop is at Day 7 → onboarding/plan problem, not motivation problem

SECTION 6: EXPERIMENT IF WEEK 1 RETENTION IS BELOW 20%

Situation: Week 1 Completion < 20% (3+ workouts in first 7 days)

If only 20% or fewer users complete 3+ workouts in Week 1, we have a critical problem. The Week 4 target (40%) becomes mathematically impossible.

First: Diagnose before experimenting

Before running any experiment, check these diagnostic questions:

Is Day 1 workout completion above 50%?

- └── NO → Problem is first impression (plan clarity or onboarding)
- └── YES → Problem is Week 1 continuation (motivation or friction)

Is check-in rate above 40%?

- └── NO → Users aren't engaging with the daily feature (UX friction)
- └── YES → Users engage but still drop off (content problem)

Are any users flagging "confused" in check-ins?

- └── YES → Plan clarity problem (language, videos, exercise choice)
- └── NO → Motivation problem (need psychology, not better instructions)

Experiment Design: "Minimum Viable Habit" Test

Hypothesis: Users fail Week 1 not because the workouts are bad, but because the bar is too high. Users who feel they've "failed" (didn't do 45-min session) stop coming back. A lower, guaranteed-success entry point would prove to users they CAN do this.

Experiment Name: Minimum Viable Habit Week

Structure:

- **Control group (A):** Receives standard Week 1 plan (30-45 min, 4-5 exercises, 4 days)
- **Test group (B):** Receives "Minimum Viable Habit" Week 1 plan

Test group (B) Week 1 plan:

- Day 1-3: 25-minute "starter" sessions (3 exercises, 2 sets each — enough to feel like a real workout)
- Day 4-7: 35-minute "growth" sessions (4 exercises, 3 sets each)
- Message framing: "Week 1 is about building the habit of showing up. Intensity picks up Week 2."
- Success bar: Complete any 3 of 7 days (not 4 of 5)

Rationale:

- Lowers psychological barrier (10 min is non-negotiable)
- Creates guaranteed early wins (anyone can do 10 min, 2 exercises)
- Shifts identity ("I'm someone who goes to the gym") before volume increases
- Based on Tiny Habits research (BJ Fogg): start smaller than feels necessary

What we measure:

- Primary: Week 1 completion (3+ days) — Group A vs Group B
- Secondary: Week 2 re-engagement — do Group B users continue?
- Safety: Week 4 completion — does lighter Week 1 still lead to Week 4?

Success criteria:

- Week 1 completion: Group B > Group A by 15%+
- Week 2 re-engagement: Group B ≥ Group A (not significantly lower)
- Week 4: Group B ≥ 30% (acceptable even if lower than standard)

What we learn:

- If B beats A: Volume reduction in Week 1 is the right call → Update V1 for all users
- If A beats B: Volume isn't the problem → Problem is somewhere else (plan clarity, motivation, videos)
- If B Week 1 is better but Week 4 is much worse: Short burst doesn't build habit → Experiment failed, try different hypothesis

Timeline: Run for 2 weeks with 20+ users per group (minimum for signal)

Backup Experiments (If Primary Doesn't Run)

Experiment B: "Accountability Buddy"

- Hypothesis: Solo exercise without external accountability is too easy to skip
- Test: Pair users with one "accountability partner" (another GymBuddy user)
- Measure: Does paired group show higher Week 1 completion?
- Risk: Adds social complexity to MVP — test after Experiment A

Experiment C: "Celebration Timing"

- Hypothesis: Celebration message after workout (current) comes too late — users need motivation BEFORE they start, not after
- Test: Send "You're about to do something great" message 30 min before planned workout time
- Measure: Does pre-workout motivation message increase Day 1 completion?
- Risk: Requires notification permissions — address UX first

SECTION 7: COMPLETE METRICS FRAMEWORK (DASHBOARD READY)

Metric Definitions for Dashboard

Metric	Category	Definition	Formula	Target	Frequency
Week 4 Completion Rate	North Star	% onboarded users who check in during Days 22-28	$\text{W4 check-ins} \div \text{Total onboarded} \times 100$	40%+	Daily (rolling)
Day 1 Workout Completion	Leading	% users who complete first workout	$\text{Post-workout check-ins Day 1} \div \text{Onboarded} \times 100$	70%+	Real-time
Week 1 Consistency (3+)	Leading	% users who do 3+ workouts in Days 1-7	$\text{Users with 3+ check-ins Week 1} \div \text{Onboarded} \times 100$	65%+	Day 7
Daily Check-in Rate	Leading	% of planned workouts with check-in	$\text{Check-ins submitted} \div \text{Planned workouts}$	65%+	Daily
Week 2 Re-engagement	Leading	% Week 1 users who return in Week 2	$\text{W2 active users} \div \text{W1 active users} \times 100$	55%+	Day 14
Habit Recovery Rate	Leading	% who resume within 3 days of miss	$\text{Returns in 3 days} \div \text{Users who missed} \times 100$	40%+	Weekly

Mid-Week Checkpoint ROI	Leading	% who work out after Layer 3 adaptation	Post-adaptation workouts ÷ Users adapted × 100	60%+	Weekly
Confidence After Day 1	Qualitative	"Very confident" after first workout	In-app survey responses	80%+	Day 1
Plan Clarity Score	Qualitative	"Completely understood" first plan	In-app survey responses	90%+	Day 1
Adaptation Usefulness	Qualitative	"Very helpful" on first adaptation	In-app survey responses	75%+	First adaptation
NPS	Qualitative	Net Promoter Score (0-10 recommended)	(Promoters - Detractors) ÷ Total × 100	40+	End of Week 4

SECTION 8: MVP CURRENT NUMBERS

Note: These will be populated as users onboard. Template shown below.

Week 1 Launch (Days 1-7)

Metric	Target	Actual	Status
Users Onboarded	20+	[LIVE]	 Tracking
Day 1 Workout Completion	50%+	[LIVE]	 Tracking
Week 1 Consistency (3+)	45%+	[LIVE]	 Tracking
Daily Check-in Rate	45%+	[LIVE]	 Tracking

Confidence After Day 1	70%+	[LIVE]	 Tracking
Plan Clarity	70%+	[LIVE]	 Tracking

Week 2 Check (Days 8-14)

Metric	Target	Actual	Status
Week 2 Re-engagement	45%+	[LIVE]	 Tracking
Habit Recovery Rate	40%+	[LIVE]	 Tracking
Mid-Week Checkpoint ROI	40%+	[LIVE]	 Tracking
Adaptation Usefulness	65%+	[LIVE]	 Tracking

Week 4 Final (Days 22-28)

Metric	Target	Actual	Status
Week 4 Completion Rate	40%+	[LIVE]	 PRIMARY
NPS	40+	[LIVE]	 Tracking
"Would you continue using GymBuddy?"	60%+	[LIVE]	 Tracking

Product

GymBuddy – An AI-powered adaptive fitness coach that helps beginner and inconsistent gym-goers build a sustainable workout habit through personalized weekly plans, daily adaptive check-ins, and mid-week plan adjustments.

1. Target Audience

Primary Users

- Beginner gym-goers (18–35 years)
- First-time gym members
- Users who have previously quit within the first month
- Working professionals with inconsistent schedules

Initial Launch Locations

Instead of targeting everyone, focus on:

- Budget gyms (₹500–₹1500/month)
- Cult.fit centers
- Apartment society gyms
- Office gyms
- College gyms

These environments have the highest concentration of users who match your

personas.

2. Value Proposition (Elevator Pitch)

"Most people don't quit the gym because they're lazy—they quit because life gets in the way. GymBuddy adapts your workouts based on your available time, energy, schedule, and health conditions so you can stay consistent and complete your first four weeks."

3. User Acquisition Strategy

Phase 1 – Personal Network

Acquire the first users through:

- Friends
- Office colleagues
- Family
- College groups

Goal:

20–30 users

Phase 2 – Local Gym Outreach

Visit:

- Local budget gyms
- Cult.fit
- Apartment gyms

Speak with beginners.

Offer:

"We're testing an AI fitness coach for beginners. Would you like early access?"

Collect sign-ups using:

- QR Code
- Google Form
- Direct website signup

Phase 3 – WhatsApp Communities

Share in:

- Apartment groups
- Gym WhatsApp groups
- Office wellness groups

Example message:

Hi everyone! 🙌 I'm building GymBuddy, an AI fitness coach that creates personalized workout plans and adapts them based on your daily schedule, energy, and health conditions. If you're starting your fitness journey or struggling to stay consistent, I'd love your feedback. Try it here: [GymBuddy Link].

Phase 4 – Referral

Every activated user receives:

Invite one friend who is starting the gym.

Simple referral:

Bring a workout buddy.

4. Onboarding Strategy

The onboarding must take **less than 2 minutes**.

Collect only essential information:

- Goal
- Experience
- Health conditions
- Available workout days
- Workout duration
- Preferred location

Immediately generate the AI workout plan.

Avoid asking unnecessary questions that increase drop-off.

5. Activation Metric

A user is considered **Activated** when they complete all of the following:



Complete onboarding



Generate their first AI weekly plan



Complete their first daily check-in

This shows they've experienced the core value of GymBuddy.

6. Success Funnel

100 Users Approached



60 Signed Up



50 Completed Onboarding



42 Generated Weekly Plan



35 Completed First Check-in



25 Completed Week 1

↓

18 Completed Week 4

Stage	Definition	Target
Approached	Users introduced to GymBuddy	50
Signed Up	Created an account	30
Activated	Completed onboarding + generated plan + first check-in	25+
Week 1 Completed	Finished first week	25+
Week 4 Completed	Finished the program	20+

8. Churn Hypothesis

Based on interviews and survey findings, users are most likely to churn because:

- Work becomes busy
- Motivation decreases
- They miss a few workouts
- They feel guilty after falling behind
- Static plans become unrealistic

9. Retention Strategy

If users miss workouts:

Instead of saying:

"You missed three workouts."

GymBuddy says:

"Looks like you've had a busy week. Let's simplify the remaining plan so you can finish strong."

This reinforces the product philosophy of **supporting users instead of punishing them**.

10. If Early Users Churn

If Week 1 retention is low, I would conduct follow-up interviews to identify the primary reason for drop-off and iterate based on the findings. Potential experiments include:

Problem	Experiment
Onboarding too long	Reduce onboarding questions and shorten completion time
Users forget workouts	Introduce personalized reminder notifications

Plans feel overwhelming	Default beginners to shorter 20–30 minute workouts
Motivation drops	Add encouraging messages and progress milestones
Users skip workouts	Improve the mid-week adaptive review to help users recover without feeling behind

STEP 7 – Risks, Safety & Reflection

Risk	Likelihood	Impact	Mitigation Strategy
AI recommends an unsuitable workout for a user's health condition	Medium	High	Collect health conditions during onboarding and re-check for pain before adapting workouts. Avoid exercises known to aggravate common conditions and clearly state that GymBuddy is not a medical substitute.
Users don't trust AI recommendations	Medium	High	Explain every adaptation. Example: <i>"Today's workout was shortened because you reported only 20 minutes available."</i> Transparency builds confidence.

Users ignore the adaptation popup because it interrupts workouts	Medium	Medium	Keep the popup to a single question: " Has anything changed since your weekly plan? " If "No," users immediately continue to their workout. Only ask detailed questions when necessary.
Users skip multiple workouts during the week	High	High	Trigger the Mid-week Adaptive Review to condense or rebalance the remaining workouts instead of expecting users to catch up.
Internet connectivity inside gyms is poor	Medium	Medium	Store the weekly workout locally. AI adaptation is requested only when the user reports changes, reducing dependency on constant internet access.

AI-generated workouts are inconsistent	Medium	High	Use structured prompts with predefined safety rules, beginner-friendly exercises, and validation logic before displaying recommendations.
Users provide inaccurate onboarding information	Medium	Medium	Allow users to edit goals, experience level, and health conditions anytime. Use weekly reviews to keep profiles updated.
Beginners perform exercises incorrectly	High	High	Include exercise images, instructions, equipment guidance, and form tips. Encourage consulting a trainer for unfamiliar movements.
Users expect GymBuddy to diagnose injuries or prescribe treatment	Low	High	Clearly communicate that GymBuddy provides fitness guidance only and advise users to seek medical care for injuries or concerning symptoms.

Diet recommendations may not suit all users	Low (V1)	High	V1 does not provide meal plans. If nutrition is added later, recommendations should remain general and avoid prescribing therapeutic diets without professional guidance.
--	----------	------	---

Honest Reflection (Updated)

Since you've introduced nutrition into V1, I'd also slightly revise your reflection.

The product decision I'm least confident about is combining adaptive workouts and nutrition guidance within the MVP. While interviews showed users wanted both, adding nutrition increases the complexity of personalization and safety. Poor nutrition advice can have a greater impact than an unsuitable workout, especially for users with medical conditions. For the MVP, I've intentionally limited nutrition to **goal-based educational guidance** (calories, protein, hydration, and sample meals) rather than highly personalized therapeutic meal plans. If early users find nutrition valuable, I would validate this further with dietitians before expanding the feature.

Initially, I planned a mandatory daily check-in where users answered questions about energy, soreness, time, and motivation before each workout.

However, after thinking through the real gym experience, I realized that this could become repetitive and interrupt users who simply want to start exercising. The additional interaction might create unnecessary friction and reduce engagement.

To address this, I redesigned the experience.

Instead of a full daily questionnaire, GymBuddy now asks a single question:

"Has anything changed since your weekly plan was created?"

If the answer is **No**, the user immediately starts the planned workout.

If the answer is **Yes**, GymBuddy asks only the relevant follow-up questions (time available, energy level, soreness, pain, location, etc.) and adapts the workout accordingly.

I believe this balances **personalization** with **speed**, but I would validate this assumption by measuring:

- Adaptation popup completion rate
- Percentage of users selecting "No changes"
- Workout start rate after the popup
- User feedback on whether the interaction feels helpful or intrusive

If users still find it disruptive, I would experiment with triggering the adaptation flow only after missed workouts or when a user explicitly requests changes.

AI Safety Guardrails (Updated)

1. Health Before Performance

User safety always takes priority over workout progression.

During onboarding, GymBuddy collects:

- Health conditions
- Injuries (if any)
- Fitness goal
- Experience level

During workout adaptation, if users report:

- Knee pain
- Shoulder pain
- Lower back pain
- Excessive soreness
- Fatigue

GymBuddy modifies or replaces exercises with safer alternatives rather than encouraging users to push through pain.

2. Beginner-First Workout Programming

GymBuddy is designed primarily for beginner and inconsistent gym-goers.

The AI:

- Recommends beginner-friendly exercises
- Avoids highly technical or advanced lifts
- Uses conservative training volume
- Provides exercise instructions and equipment guidance
- Encourages gradual progression rather than aggressive overload

3. Nutrition Safety

GymBuddy provides **goal-based nutritional guidance**, not medical nutrition therapy.

Recommendations are personalized using:

- User goal (Muscle Gain / Fat Loss / General Fitness)
- Age
- Gender
- Height & Weight
- Activity level
- Dietary preference (Vegetarian / Non-vegetarian / Vegan)
- Health conditions

The AI may recommend:

- Daily calorie range
- Protein intake
- Hydration goals

- Meal timing suggestions
- Example meal ideas

However, GymBuddy **does not**:

- Prescribe therapeutic diets
- Recommend supplements or medications
- Create nutrition plans for medical treatment
- Replace a registered dietitian or doctor

If a user has diabetes, kidney disease, severe food allergies, eating disorders, pregnancy, or any complex medical condition, GymBuddy displays a disclaimer recommending consultation with a qualified healthcare professional before following nutritional advice.

4. Transparent AI Decisions

Whenever GymBuddy adapts a workout or nutrition recommendation, it explains why.

Example (Workout):

"You reported low energy and only 20 minutes available today, so I've shortened today's workout while keeping the primary compound movement."

Example (Nutrition):

"Your protein recommendation has been increased because your goal is muscle gain and you're consistently completing your weekly workouts."

This transparency helps users understand and trust the AI's reasoning.

5. Conservative Progression

GymBuddy only increases training difficulty when:

- Previous workouts were completed successfully
- Recovery is adequate
- No pain or injuries are reported
- User feedback indicates the current plan feels manageable

Otherwise, the AI maintains or reduces workout intensity to prioritize consistency and safety.

6. Medical Boundaries

GymBuddy is an AI fitness coach, **not a medical device**.

It does not:

- Diagnose injuries
- Prescribe rehabilitation programs
- Recommend exercising through pain
- Replace professional medical advice

If users report symptoms such as:

- Chest pain
- Severe dizziness
- Loss of consciousness
- Difficulty breathing
- Sudden swelling or acute injury

GymBuddy immediately advises them to stop exercising and seek medical

attention.

7. User Control

Users remain in control of their fitness journey.

They can update their:

- Fitness goals
- Health conditions
- Dietary preferences
- Workout schedule
- Available workout time

This ensures recommendations remain relevant as their circumstances change.